



Department of Information Technology

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch : V Semester-B. Tech Information Technology

Course Code & Title : IT3501 Full Stack Web Development

Name of the Faculty member (s) : S. Sakkaravarthi, AP/IT

Innovative Practice Description

- **Unit / Topic** : Unit 2 / Basics of Node.js & Installation
- **Course Outcome** : CO2
- **Topic Learning Outcome:** TLO5
- **Activity Chosen** : Flipped Class Room
- **Justification:**
 - Flipped classroom is a teaching method where the usual approach to classroom instruction is reversed. In this model, students learn basic concepts on their own by using materials such as readings and pre-recorded video lectures.
 - The flipped classroom method is adopted to teach the topic Basics of Node.js and Installation in an engaging and learner-centered manner.
- **Time Allotted for the Activity:** 40 Minutes
- **Details of the Implementation:**
 - Prior to the class, students were provided with curated learning resources, including videos, short notes, and a step-by-step installation manual for Node.js.
 - Students were instructed to go through these materials at home to gain familiarity with concepts such as what Node.js is, how it works, its features, its uses, and the installation process.
 - During the actual class session, students were grouped into teams of two or four and given 10 minutes to discuss the topic they had studied at home using the provided materials, as shown in Fig. 1.
 - After the discussion, each team presented their understanding of the Node.js basic concepts and installation steps to the entire class as shown in Fig 2.
 - The instructor guided the discussions, addressed misconceptions, and demonstrated best practices wherever necessary.

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO2	3	2	3	2	2	3

(1 – Low

2 – Moderate

3 – High)



• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	3	2	3	2	2	3
Justification for correlation	Students gain a solid understanding of the fundamental concepts of Node.js, its architecture, and installation steps, enabling them to apply core computing principles effectively.	Students are able to install Node.js, verify the runtime, configure environment settings, and execute basic JavaScript programs from the command line.	Students develop the ability to create simple server-side programs and run them using Node.js, forming a foundation for advanced back-end development	As the student engaged individually in pre class study , their individual programming confidence and self-learning capability improved	Students participate in group discussions, peer explanations, and presentations, enhancing their communication skills while discussing technical concepts.	The basic Node.js skills learned through this activity help students apply their knowledge to real-world problems

• **Images / Screenshot of the practice:**



Fig 1: Students discussion with their team members Fig 2: Rashika and Valli share their Presentation

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Students expressed that the activity helped them evaluate their understanding of Node.js basics more clearly.
- Many felt that reviewing materials beforehand made it easier to recall the installation steps during the class.



- Students shared that this practice simplified their understanding of how to set up node.js server and basic programs.

❖ ***Benefit of the practice:***

- The activity helped students clearly grasp key ideas such as the purpose of Node.js, how to install it, and how to verify the runtime using the terminal.
- The flipped classroom approach was especially helpful for slow learners, as it allowed them to revisit the installation videos and guides at their own pace.
- Students gained confidence in executing their first Node.js script and learned how to troubleshoot common installation issues.

❖ ***Challenges faced in implementation:***

- Some students came to class without completing the required pre-learning tasks.
- Creating flipped classroom content required greater preparation time compared to conventional teaching.

Provided additional clarification on the basic concepts of Node.js during the next class.

References:

- https://www.tutorialspoint.com/the_full_stack_web_development/index.asp
- <https://www.udemy.com/course/the-full-stack-web-development/>

Signature of Faculty Member

HOD